

Note: I have put key words in **bold** in the text below. You may look these up (e.g. in the reading, or on Wikipedia) for help or context on a given problem.

I. Combinatorial Circuits

Design **combinatorial circuits** with the following input-output profiles:

1. NAND

IN 1	IN 2	OUT
0	0	1
0	1	1
1	0	1
1	1	0

2. Reconstructing OR

IN 1	IN 2	OUT
0	0	0
0	1	1
1	0	1
1	1	1

<< For this problem, do not use OR-gates.

DeMorgan's Law is useful here!

3. Splitting outputs

IN 1	IN 2	OUT1	OUT2
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

4. XOR (exclusive OR)

IN 1	IN 2	OUT
0	0	0
0	1	1
1	0	1
1	1	0

5. Combining circuits

IN 1	IN 2	OUT1	OUT2
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Hint: you might think about the circuits here as systematically analogous to equations in propositional logic, and the input-output tables as analogous to truth-tables. In many cases, you can use your knowledge of logic to work backwards to find the appropriate circuit.

II. Functions

6. Consider the function $m(\cdot)$ defined by the following **equation**:

for any integers x, y , and z : $m(x, y, z) = 3 + (x \cdot y \cdot z)$

- Describe the function with an English sentence using each of the terms “argument”, “value”, “apply”, and “return” correctly.
- Describe the function with another English sentence using the phrase “maps ... to”.
- How many **places** does this function have?

7. Consider the function $k(\cdot)$ that takes two integers x and y as arguments, and returns the **successor** of their product as its value. Define this function with an equation.

8. This is a problem about **function composition**. Let $s(\cdot)$ be the **successor function**. Let $a(\cdot)$ be the **addition function**. Suppose that $p()$ is a 2-place function composed of successor functions and addition functions. Given that $p(1, 2) = 5$, provide two possible definitions of $p()$.

9. Consider the function $j()$ that maps two integers x and y to their product, so long as both x and y are equal to or greater than 0, and equal to or less than two. For any other arguments, it is undefined.

- Define $j(\cdot)$ with an equation.
- Define $j(\cdot)$ with a table.

This is a problem about **partial functions**.

10. The **Boolean numbers** are 0 and 1, where 0 and 1 are considered to be the “opposites” of one another. A **Boolean function** is a function which has only Boolean numbers as arguments and values.

- Using one equation, define a Boolean function which maps any Boolean number to its opposite.
- Provide two different definitions of the Boolean function which maps any Boolean number to itself. This is called the **identity function**.
- Provide two different definitions of the Boolean function which maps any Boolean number to the number one. This is called a **constant function**.

III. Representations

11. Represent the numbers zero through ten in **binary**.
12. Invent a new **base-6 counting system**, with all new symbols for the 6 digits (including zero!). List your basic symbols and their meanings. Then represent the number thirty-two in this system.
13. a. Add the binary number 1111 to the binary number 1.
b. Add the binary number 1011 to the binary number 11001.
Use one of the methods of addition from the Petzold reading; show your work!
14. Define a function f which maps the symbol “○” to the number zero and the symbol “●” to the number one. Use a table or an equation.
15. Suppose we use the symbols in Problem 14 to form a base-2 counting system. Let’s call these symbols “dots”. A dot can be black or white. Using an equation, define a 3-place function g which takes any three dots in a row and maps them onto the numerical value of that string of dots in that base-2 system. You will have to use the function in Problem 14 as part of the definition.

So, for example, your function should predict that

$$g(\circ, \circ, \bullet) = \text{one} \qquad g(\circ, \bullet, \circ) = \text{two} \qquad \text{and so on...}$$

Your answer should take the form:

$$\text{“for any series of three dots } x, y, z: g(x, y, z) = \dots\text{”}$$

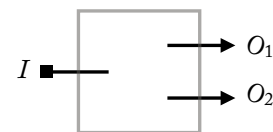
Hint: there are two things going on in Problem 15. It asks you to use your knowledge of functions to define the numerical values of symbols in a base-2 counting system. The method of computing binary values described by Petzold will be of help here. But it is also asking you to keep track of the difference between a *symbol* and its *content*, in this case the difference between “●” and *the number one*. Remember that only numbers, but not symbols, can be added and multiplied! At a deeper level, this problem asks you to give a partial **semantics** for a 2-place representational system. The science of formal semantics extends this kind of analysis to the domain of language.

Challenge problem: define a function which maps strings of 1’s and 0’s of *arbitrary length* to their numerical values. This is the general problem of semantics for binary.

IV. Synthesis

16. Let $tal(\cdot)$ be a function which maps strings of tally 1’s and 0’s to their numerical values. For example, $tal(0)=\text{zero}$, $tal(1)=\text{one}$, $tal(00)=\text{zero}$, $tal(01)=\text{one}$, $tal(11)=\text{two}$, and so on.
Consider a machine M which, on any one occasion, takes as input a single atomic symbol I , and produces two atomic symbols as output, O_1 and O_2 , such that, in any context, $successor(tal(I)) = tal(O_1O_2)$. Design a combinatorial circuit for machine M , with the schematic form below.
17. Let $bin(\cdot)$ be a function which maps strings of binary 1’s and 0’s to their numerical values. For example, $bin(0)=\text{zero}$, $bin(1)=\text{one}$, $bin(00)=\text{zero}$, $bin(01)=\text{one}$, $bin(10)=\text{two}$, $bin(11)=\text{three}$, and so on.
Consider a machine N which, on any one occasion, takes as input a single atomic symbol I , and produces two atomic symbols as output, O_1 and O_2 , such that, in any context, $successor(bin(I)) = bin(O_1O_2)$. Design a combinatorial circuit for machine N .

Hint: these two problems look complex, but they are just asking you to apply the intuitive concept of **computation** we have been discussing in the course so far. First figure out what table of input and output symbols would satisfy the equations given. Only then solve for the underlying circuit.



Schematic for M and N